

TrES-5b

R-1166

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190428 · created 2026-07-31T02:52:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458601.9217 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.022
Transit depth (Rp/Rs)^2	3.42 +/- 0.81 [%]
Orbital Inclination (inc)	85.12 +/- 1.74
Airmass coefficient 1 (a1)	1.092 +/- 0.047
Airmass coefficient 2 (a2)	-0.057 +/- 0.023
Scatter in the residuals of the lightcurve fit is	4.27 %
Best Comparison Star	#2 - [362, 359]
Optimal Aperture	3.29
Optimal Annulus	8.08
Transit Duration (day)	0.0795 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.022 vs 0.142 (deviation 1.78σ)
Duration fit vs published	0.0795 d vs 0.0758 d (deviation 0.39σ)
Mid-time O–C	1.3 min
Depth SNR	7.6
Residual scatter	4.27 %

Light curve

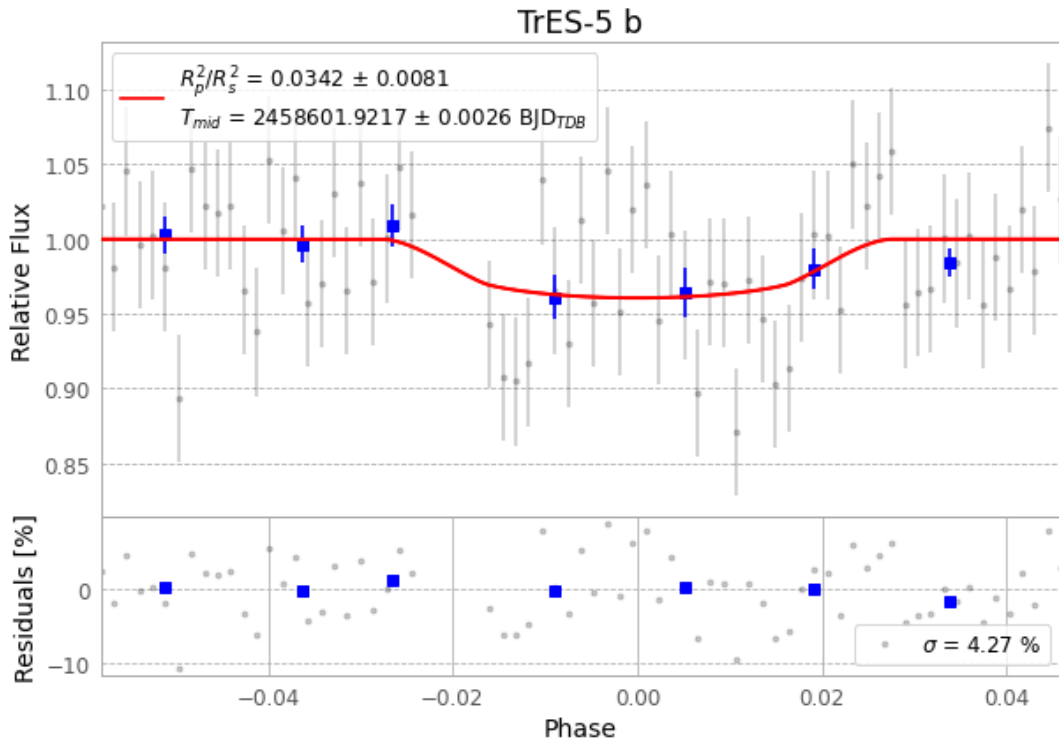


Figure 1 — FinalLightCurve_TrES-5 b_2019-04-28.png (SHA-256 3b3d6366ec2799fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 115], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 23b852eb898b49c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), TRES-5190428080312.FITS → TRES-5190428114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190428.log (96b49461317f...), FinalLightCurve_TrES-5 b_2019-04-28.png (3b3d6366ec27...), FinalLightCurve_TrES-5 b_2019-04-28.csv (a9380d2e9b03...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:52Z.